

NeuroSpark

Stage 1.5 Viability Assessment & Detailed Go-Forward Plan

A biologically-inspired spiking neural network with Active Inference — is it worth continuing, and if so, how?

VERDICT: GO — but reframe and narrow the scientific claim.

Prepared 22 June 2026. Based on full inspection of the existing Stage 1 codebase plus a 24-source, adversarially-verified deep-research pass on novelty, data access, scientific validity, publishability, and wetware feasibility.

1. Executive Verdict

Continuing is worth it — if you stop calling it a connectome model and start calling it a controlled comparison of Active Inference vs. Reinforcement Learning on a real brain connectome. That specific conjunction is the one thing in this project that the 2024–2026 literature has *not* already done. Everything broader than that is taken.

- **Do this:** Finish a tight Stage 1.5 *software* study — real FlyWire subgraph, a genuine variational/expected-free-energy implementation, a PPO/Q-learning baseline, and topology + coding + epistemic ablations with 50 trials and statistics. Target a workshop / specialized venue. Realistic effort: **~3–6 months part-time**.
- **Defer this:** Stage 2 *wetware* (living neurons / slime mold on an MEA). It is a 12+ month, wet-lab-and-funding-scale undertaking with poor return for a solo student right now.
- **Fix this first:** The current 'free energy' is a Mean-Squared-Error stand-in, not variational free energy. Until that is replaced, the headline 'Active Inference' claim is not technically earned and a reviewer will catch it immediately.

In one line: **the engineering is solid, the vision is over-claimed, and the rescue is to shrink the claim to the one defensible niche and execute it rigorously.**

2. What You Have Actually Built (Reality Check)

Stage 1 is a real, running closed-loop agent — a genuine achievement. But several headline claims in the README do not match the code on disk. Naming these now prevents a reviewer (or a faculty sponsor) from finding them later.

Claim in README / docs	What the code actually does	Impact
"FlyWire / fly-brain connectome"	Synthetic Watts–Strogatz small-world graph (brain/connectome_loader.py). The file is just <i>named</i> flywire_subset_1k.npz.	The entire publication pitch rests on <i>real</i> connectome data. This is the #1 gap.
"Variational / Expected Free Energy"	Plain MSE prediction error (inference/free_energy.py: $VFE = MSE(state) + (reward\ error)^2$). No KL, no epistemic value.	Without true VFE/EFE the 'Active Inference' label is unearned.
"Learns to navigate"	Reaches the goal (best 8 steps on 5×5; 145 episodes logged), but avg_free_energy logs as 0.0 every episode and EFE spread is often <0.02 (flat).	The core scientific signal (free-energy reduction) is not even being measured.
"Planning via imagined futures"	Implemented, but README's own "What's Left to Do" marks EFE differentiation and forward-model accuracy as unresolved P0 bugs.	Planning works mechanically but may not carry real signal yet.

Bottom line: you have an excellent *engineering scaffold and demo*. You do not yet have a *scientific result*. Stage 1.5 is exactly the work that converts the former into the latter.

3. Deep-Research Findings

Synthesized from 24 primary sources, 116 extracted claims, 25 adversarially verified (24 confirmed). Citations are listed in §8.

3.1 Novelty & prior art — the niche is narrow but real

- **Broad idea is taken.** Connectome-constrained embodied agents already exist: **FlyGM** (arXiv 2026) instantiates the whole adult *Drosophila* connectome as a graph controller for MuJoCo locomotion and beats rewired/random/MLP baselines on sample efficiency; **Lappalainen et al.** (Nature 2024) built a connectome-constrained optic-lobe network matching real activity across 26 studies; **NeuroMechFly v2** (Nature Methods 2024) runs a connectome-constrained closed-loop embodied fly.
- **Active Inference on grids is taken.** **FEPS** (PLOS One 2025) derives a policy by minimizing expected free energy on a partially-observable grid; 'Active Inference vs. RL' is a well-populated debate (Friston 2009; Sajid et al. 2021).
- **The conjunction is open.** Every connectome controller above uses *reinforcement learning* — none uses active inference / EFE. So '**Active Inference vs. RL on a real connectome**' is **genuinely unoccupied**. That, and only that, is your defensible novelty.

Implication: You may not claim 'first connectome-based agent.' You may claim 'first head-to-head test of whether an active-inference controller exploits real connectome structure differently than an RL controller does.'

3.2 FlyWire data is freely available (no special access)

- **Bulk:** Zenodo v783 (DOI 10.5281/zenodo.10676866, CC-BY-4.0) — proofread_connections_783.feather with directed pre/post IDs, synapse-count weights, and per-neurotransmitter averages.
- **Interactive:** Codex (codex.flywire.ai) — browse/export directed edges with synapse counts, neuropil, predicted NT.
- **Programmatic:** CAVEclient (pip) after a free Google CAVE login; stable LTS versions 630 & 783; cell-type annotations (Schlegel et al.) let you pick a *functionally coherent* subgraph instead of an arbitrary cut.

Gotchas to budget for: (1) NT signs are *classifier predictions* (~87–94% acc.), not ground truth — state this in the paper; (2) weights need summing across multiple per-neuropil rows per pair, above a 5-synapse threshold; (3) NT columns are probabilities → argmax to a sign (ACh excitatory; GABA/Glut inhibitory); (4) files are large (~GB) — process in chunks.

3.3 Scientific validity — defensible, with two real risks

- **Supports the hypothesis:** Fly topology is genuinely distinctive (small-worldness ~141, rich-club ~30% of neurons) — nothing like a Watts–Strogatz graph — so a 'biological vs. small-world vs. random' ablation tests something real. FlyGM already shows real topology gives a measurable inductive-bias advantage.
- **Risk #1 (subsampling):** a ~1,000-neuron cut is <1% of the 139k-neuron brain. Reviewers will ask whether it preserves the rich-club/recurrence that supposedly causes the advantage. If you cut arbitrarily, you may get a null result. **Mitigation:** select a coherent sensorimotor pathway via cell-type annotations.
- **Risk #2 (MSE ≠ free energy):** the active-inference claim collapses unless you implement true VFE/EFE with an epistemic term. Non-negotiable.

3.4 Publishability — workshop / specialized-venue bar

Top-tier connectome modeling (Nature-family) sets a fidelity bar a 1,000-neuron toy-grid SNN cannot meet. But scoped as a methods comparison, this is a credible **workshop or specialized-journal** paper. Realistic targets:

Venue	Type	Fit
arXiv / bioRxiv	Preprint	Immediate — stake the claim once results exist

Venue	Type	Fit
IWAI (Int'l Workshop on Active Inference)	Workshop	Very high — exactly this niche
NeurIPS / ICLR NeuroAI workshops	Workshop	High — bio-inspired AI audience
ALIFE / Artificial Life	Conf./Workshop	High — embodied + bio substrate
Frontiers in Comp. Neuroscience / eNeuro	Journal	Medium–High — if ablations are clean

To clear the bar: pre-registered ablation design, adequate statistical power (the 50-trial plan is good), an honest prior-art section citing FlyGM / Lappalainen / NeuroMechFly / FEPS, and an explicit statement that you replaced the MSE proxy with real free energy.

3.5 Wetware Stage 2 — defer

DishBrain (Cortical Labs) required cultured cortical neurons on a high-density CMOS MEA, sterile culture + incubation, custom closed-loop electrophysiology, and a multidisciplinary team. FinalSpark offers remote cloud access to living-neuron MEAs (lowers the hardware bar) but viability, interfacing and latency stay hard. Slime mold is cheap and dorm-friendly but **is not a spiking system** and does not map onto your SNN/active-inference framework. For a solo student without a wet lab this is a separate, externally-funded project — not the next step. (This was the weakest-evidenced research angle; re-research before any commitment.)

4. The One Thesis To Build Around

“Does an Active-Inference controller exploit a real *Drosophila* connectome's structure more sample-efficiently than reinforcement learning — and does biological topology matter versus random and small-world controls?”

Every task below exists to answer that sentence with statistics. If a piece of work does not serve it, cut it.

5. Stage 1.5 — Detailed Go-Forward Plan

Five phases. Each has a concrete deliverable and a verification gate; do not start the next phase until the gate passes. Effort assumes part-time solo work.

Phase 0 — Repository hygiene & honesty pass (½–1 day)

- Add a real **.gitignore** (venv/, __pycache__/, *.mp4, data/episode_feedback/, *.npz, *.pdf build artifacts). The repo currently commits a 1.1 GB venv.
- Rename the synthetic connectome file/variables away from 'flywire' until the real one lands (truth-in-labeling).
- Fix `avg_free_energy` logging so the FE trace is actually recorded per step — you cannot study what you do not measure.
- Delete foreign junk (see §7).

Phase 1 — Real FlyWire connectome (3–5 days)

- Pull a coherent ~1,000-neuron subgraph along a sensorimotor pathway using Codex/CAVEclient + Schlegel cell-type annotations (not an arbitrary BFS).
- Build a signed, directed, synapse-count-weighted matrix: sum rows per pair, threshold at 5 synapses, $\text{argmax NT} \rightarrow \text{excitatory/inhibitory sign}$.
- Map onto the existing Brian2 LIF network in place of Watts–Strogatz.
- **Gate:** spike raster shows structured, non-random bursting; degree/weight distributions match published FlyWire stats (Figure 1 of the paper).

Phase 2 — Real (Expected) Free Energy (1–2 weeks)

- Replace MSE with a proper generative model (A: $P(o|s)$; B: $P(s'|s,a)$; C: preferences; D: prior) and compute $\text{VFE} = \text{complexity} - \text{accuracy}$, $\text{EFE} = \text{risk} + \text{ambiguity} - \text{epistemic value}$.
- Keep the interface identical so encoder/decoder/loop are unchanged.
- **Gate:** free energy provably decreases within and across episodes; EFE differs across actions (spread $\blacksquare$ 0.02). This retires the two open P0 bugs.

Phase 3 — RL baseline + ablation grid (2–4 weeks, compute-bound)

Same environment, same encoder/decoder. Run 50 trials $\times$ each condition with fixed seeds:

Condition (axis)	Tests
FlyWire + Active Inference (full model)	The headline system
PPO / Q-learning baseline	Is Active Inference actually better/worse?
Random connectome	Does biological topology matter at all?

Condition (axis)	Tests
Watts–Strogatz small-world	Is it just small-worldness, or specifically fly wiring?
Rate-coded RNN (no spikes)	Does spike timing carry information?
Epistemic term off	Does curiosity help?

Gate: results saved to a single results file; pipeline reproducible from one command.

Phase 4 — Statistics, figures, write-up (2–4 weeks)

- Metrics: steps-to-goal, sample efficiency (area under learning curve), final free energy. Report mean ± SEM, t-tests, and Cohen's d — with a multiple-comparisons correction.
- Five figures: connectome stats; learning curves (all conditions); free-energy dynamics; spike raster; ablation bar chart with significance.
- 6-page paper: honest prior-art section is mandatory. Preprint to arXiv/bioRxiv, then submit to IWAJ or a NeuroAI workshop.

Total realistic effort: ~3–6 months part-time (data: days; FE + baseline + ablations: weeks, mostly compute; stats + writing: weeks).

6. Risks & Mitigations

Risk	Likelihood	Mitigation
MSE-FE not replaced → 'Active Inference' claim rejected	High if skipped	Phase 2 is non-negotiable and gated.
<1% subgraph destroys the topology effect → null result	Medium	Select a coherent pathway by cell type; report subgraph graph stats.
FlyGM/related work pre-empts novelty further	Medium	Lead with the AIF-vs-RL axis; re-run a lit check before submission.
Brian2 too slow for 50× trials × 6 conditions	Medium	Cache connectome; reduce sim window; consider Brian2 C++ standalone / GPU.
Scope creep back into wetware	Medium	Freeze Stage 2 until Stage 1.5 is submitted.

7. Junk Identified For Deletion

These are unrelated to NeuroSpark (they belong to an unrelated Rust tool, 'sentrux') or are agent scratch leftovers. None is referenced by any project code.

- **sentrux.exe** (24 MB) — foreign binary, untracked.
- **.sentrux/** — sentrux config/baseline, untracked.
- **install.sh** — sentrux installer script, untracked.
- **brain/8f96c6c2-...-/scratch/** — UUID-named agent scratch test files.

Kept on purpose: venv/ (your environment — gitignore it, don't delete), data/ (your training results), the planning PDFs, and generated PNG/MP4 artifacts.

8. Key Sources

1

FlyGM — connectome-as-controller for embodied locomotion (arXiv:2602.17997, 2026)

2

Lappalainen et al. — connectome-constrained network predicts neural activity (Nature 2024, s41586-024-07939-3)

- 3 NeuroMechFly v2 — connectome-constrained closed-loop embodied fly (bioRxiv 2023.09.18.556649)
- 4 FEPS — expected-free-energy policy on a partially-observable grid (PLOS One 2025, PMC12410762)
- 5 FlyWire connectome — whole-brain connectivity & statistics (Nature 2024, s41586-024-07558-y)
- 6 FlyWire neurotransmitter predictions (Nature 2024, s41586-024-07968-y)
- 7 FlyWire data — Zenodo v783 bulk download (DOI 10.5281/zenodo.10676866)
- 8 Codex FlyWire explorer FAQ (codex.flywire.ai/faq)
- 9 CAVEclient tutorial & materialization docs (seung-lab FlyConnectome; caveclient.readthedocs.io)
- 10 IWAI — International Workshop on Active Inference (iwaiworkshop.github.io)
- 11 NeuroAI workshop (neuroai-workshop.github.io)
- 12 Cortical Labs / DishBrain background (en.wikipedia.org/wiki/Cortical_Labs); FinalSpark Neuroplatform (finalspark.com/neuroplatform)

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